

Epigenetic Language Models: Lifelong Learning by Intercalating Knowledge into Structural Zeros

Free-Slot Adaptation on a real 8B ternary MoE

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AI-assisted draft. revised through repeated adversarial review; the experiments were run and checked by the authors. The synthetic sections are a **dense fp32 mechanism study**; the real-model section evaluates ternary patches on a real 8B ternary MoE on a 40-fact task (single seed). The pre-registered multi-session epigenesis experiment is in progress; the personal-model use case (§7) remains an *untested proposal*. Code: github.com/chutesai/epigenesis.

Abstract

We study Free-Slot Adaptation (FSA) through a dense fp32 synthetic mechanism assay and a real 8B ternary mixture-of-experts model. FSA freezes the base and writes acquired knowledge into structural zeros; removing the patch restores the original model exactly. On a 40-fact task, ternary Bop recalls all 80 held-out test items from 812,420 filled slots (a 0.9 MB patch counting positions), with bit-exact revoke, at +0.027 nats/token general-text damage; regularized LoRA recalls 78/80 with no measurable damage. This two-item recall difference on one seed is not a significant win: ternary FSA offers a smaller patch that preserves the packed format, while LoRA with a local penalty has lower damage. In the synthetic fp32 assay, tuned LoRA remains both more capable and more storage-efficient; masked retention is exact by construction, and the always-on soft variant degrades with load. All reported experiments are single-seed. Multi-session acquisition of concepts, the intended personal-model application, remains untested in a pre-registered experiment now in progress.

1 Introduction

Adding knowledge to a trained model usually means adding parameters—a LoRA adapter [Hu et al., 2022], an extra expert, or a full fine-tune that risks forgetting. We ask a narrower question: an extreme-quantized model has already paid for a pool of weight slots held at zero; can those zeros double as a place to put new knowledge?

Free-Slot Adaptation (FSA). Store a model in format Φ (here **pair8**: ternary weights, at most two of every four aligned pairs non-zero). Let $\mathcal{S}_0$ be the non-zero support and $\mathcal{Z}_0$ the zero reserve. To learn a new domain d : freeze $\mathcal{S}_0$ and shared parameters, train a disjoint slice $\mathcal{R}_d \subseteq \mathcal{Z}_0$, and store a mask M_d . At inference the base uses $\mathcal{S}_0$; domain d uses $\mathcal{S}_0 \cup \mathcal{R}_d$. The mechanism is Mallya and Lazebnik [2018]’s; the only new element is that the free weights come from the compression format rather than a pruning pass.

What we claim. The synthetic assay characterizes exact masked preservation, bounded memorization, and the cost of a dense fp32 reserve. The real-model experiment (§6) shows that a legal ternary reserve carries facts and revokes exactly. Bop matches, perhaps slightly exceeds, LoRA’s recall on this single-seed task, from a much smaller patch, but at measurable general-text damage; regularized LoRA remains the lower-damage method. We do not claim an overall advantage over LoRA or validated multi-session learning.

Contributions. (1) Quantization zeros as a continual-learning reserve, with accounting for positions, values, scales, and inference format. (2) A synthetic mechanism study: exact masked

retention, a measured reserve maximum, a degrading soft variant, and an indicative, uncontrolled fp32 LoRA comparison that favors LoRA. (3) Real 8B results: legal ternary patches, latent-free and learn-then-prune writers, a KL floor (plausibly from routing near-ties; a hypothesis), and a recall–damage–storage comparison. (4) A revocable-patch contract and a pre-registered, still untested multi-session concept experiment (§7).

2 Background and related work

Mask-based continual learning. PackNet [Mallya and Lazebnik, 2018] freezes kept weights and trains freed ones, recovering a task by its mask; Piggyback [Mallya et al., 2018] *learns masks over fixed weights* (it does not refill zeros); HAT [Serra et al., 2018] uses hard task-attention and reports *reduced*, not zero, forgetting; Supermasks [Zhou et al., 2019] make the mask itself the trainable object; Progressive Networks [Rusu et al., 2016] add and could drop columns. FSA is PackNet with the free support supplied by the quantization format. **Quantized continual learning.** Ada-QPackNet [Pietroni et al., 2023] and TinySubNets [Pietroni et al., 2025] already combine PackNet-style capacity reuse with quantization; SPDF [Thangarasa et al., 2023] reopens zeroed pretrained weights for dense fine-tuning. These sharply narrow our novelty; our distinct element is a reserve *pre-addressed by the deployed format* and composed by union. **Parameter-efficient and continual adapters.** LoRA [Hu et al., 2022], QLoRA [Dettmers et al., 2023], QA-LoRA [Xu et al., 2024]; Biderman et al. [2024] find LoRA learns and forgets less than full fine-tuning (a possible rank explanation, not a theorem); O-LoRA [Wang et al., 2023] does continual learning in orthogonal LoRA subspaces. **Sparse / composable adaptation.** Diff pruning [Guo et al., 2021] and composable sparse fine-tuning [Ansell et al., 2022]; union of *disjoint* sparse supports is the easy case of the latter. **Editing, memory, unlearning.** MEMIT [Meng et al., 2023] edits facts in place and can override; FSA adds paths that can suppress or correct outputs while enabled but cannot edit a base weight; GRACE [Hartvigsen et al., 2023] and WISE [Wang et al., 2024] keep deletable, routed side-memory for sequential edits while freezing the base; SISA [Bourtole et al., 2021] is exact unlearning by shard isolation; TOFU [Maini et al., 2024] benchmarks fictitious-fact forgetting. **Personalization and retrieval.** OPPU [Tan et al., 2024] attaches a per-user PEFT module (removed by user identity, not per record); LaMP [Salemi et al., 2024] is a personalization benchmark with retrieval baselines; RAG [Lewis et al., 2020] and kNN-LM [Khandelwal et al., 2020] store facts externally and delete them by dropping a record—the baseline any “delete a fact” claim must beat. **Low-bit models.** BitNet b1.58 [Ma et al., 2024, 2025] and LittleBit [Lee et al., 2025]; our assay methodology follows knowledge-capacity scaling [Allen-Zhu and Li, 2024]. Bop [Helwegen et al., 2019] optimizes binary weights without latent real weights; we extend its gradient-memory rule to ternary birth and death (§6.3).

3 Method

3.1 Format and reserve

In `pair8`, weights form aligned blocks of eight (four pairs); each weight is ternary times a scale, and at most two pairs per block are non-zero. A block thus has 417 reachable states ($1 + 4 \cdot 8 + \binom{4}{2} 8^2$), i.e. $\log_2 417/8 \approx 1.088$ bits/weight asymptotically (1.125 at a fixed 9 bits/block), and is $\geq 50\%$ zero *by construction* (ternary alone guarantees no such fraction). **Writable reserve.** Zeros inside already-active pairs can be filled while preserving pair occupancy and the codebook. At two active pairs, opening a third pair is illegal, but filling a zero member of an active pair is legal. Thus the writable in-codebook reserve is a subset of the full zero support, not necessarily empty in a full-occupancy block. Opening additional pairs requires spare pair occupancy or a separate patch

outside the codebook. Filled positions and signs must still be stored for revocation; the joint rate of 1.088 bits/weight is not the marginal price of a filled slot.

3.2 Adaptation and the soft variant

Hard FSA, domain d : freeze $W[\mathcal{S}_0]$, all shared parameters, and—crucially for the real format—the discrete pair occupancy and the per-row scales; train only $W[\mathcal{R}_d]$ ($\mathcal{R}_d \cap \mathcal{S}_0 = \emptyset$) with the base live; store M_d . Inference: base $\rightarrow \mathcal{S}_0$; domain $\rightarrow \mathcal{S}_0 \cup \mathcal{R}_d$ (needs a task ID). **Soft FSA** removes the task ID: freeze the base, train the reserve with a KL penalty keeping base-input outputs unchanged, and run unmasked. This is Learning-without-Forgetting [Li and Hoiem, 2017] on a frozen disjoint support, with full base-input replay—a substantial rehearsal resource we note explicitly.

3.3 Honest accounting

Plain ternary FSA adds **zero new logical parameters**: the element count is unchanged. It fills only zeros inside active pairs, so a merged patch uses the same packed tensors and adds no tensors or inference compute. This claim depends on preserving pair occupancy and the serving format; opening new pairs or using a separate patch can add work. The packed serving path was not benchmarked (§6). Storage is not free: revocation needs positions and signs in a stored diff, or the original pack. The learned-block-scale arm adds parameters and storage and is outside these packed-format claims. A full per-domain mask would cost one bit per weight; sparse position coding avoids that cost for sparse patches. Frozen row scales constrain each ternary slot to $\{-\alpha, 0, +\alpha\}$, a capacity restriction. Actual position-inclusive patch sizes are reported in §6; hypothetical trit-only accounting is insufficient.

4 Theory

Base invariance (by construction, narrow). Restrict the forward to M_0 . If adaptation changes only $W[\mathcal{R}_d]$ with $\mathcal{R}_d \cap \mathcal{S}_0 = \emptyset$ and freezes all shared state, the masked base output is unchanged—an invariance of freezing, which is why measured “retention = 1.000” is a unit test that the mask was applied (it would hold even if the reserve learned nothing). *Three subtleties:* (i) it is the *masked* base; with the reserve enabled, the base behavior is not preserved (we measure strongly negative unmasked retention). (ii) In the real **pair8** format the decoded weight is $\alpha \cdot \text{trit}$ and a quantizer that recomputes pair occupancy or scales could change the base; invariance requires freezing base support, occupancy, and scales, as in §6. (iii) The soft variant is outside this argument entirely.

Reserve capacity is accounting, not a law. We do not vary ρ or the per-slot cost, so the curve is a memorization curve, not a test of a capacity expression. We therefore report $K_{\text{reserve}} \approx \kappa_{\text{adapt}} B_{\text{patch}}$ with B_{patch} the measured patch *bits* and κ_{adapt} an *observed* knowledge-bits-per-stored-bit efficiency conditional on base, support, scales, and training budget—and note it has no term for the interference decline we plot.

5 Experiments

Synthetic knowledge-capacity assay: disjoint random fact sets $A \rightarrow B$ (key→label, $V=1024$, so each fact is 10 bits of arbitrary lookup), $K = N \log_2 V - \sum \text{CE} / \ln 2$. A 917k-parameter MLP ($\rho=0.5$) is loaded with 128k base facts (base recall only 0.744—“saturated” is relative), then adapted. **The reserve is a dense fp32 mask:** this is why everything here is a mechanism demonstration. Single seed; LoRA uses a fixed scale 2.0 (not the standard α/r of Hu et al. [2022]) with its learning rate tuned down per load to compensate; it also uses AdamW [Loshchilov and Hutter, 2019] weight decay 0.01 (vs FSA’s 0) and sees a dense base. Full fine-tuning and rank-16 LoRA start with base recall about 0.999–1.000; LoRA ranks 64, 256, and 1024 start at about 0.64. FSA sees a 50%-masked base with recall 0.744, so the comparison is indicative, not controlled.

Retention and reserve (Fig. 1). Masked retention is exactly 1.000; full fine-tuning goes below random. The reserve fills to 490.6 kbits and then declines.

LoRA (Fig. 2). With its learning rate tuned, LoRA learns facts well and at $r \geq 256$ acquires *more* than FSA’s maximum; counted honestly in fp32 it is also *more* storage-efficient. The smaller ternary patches measured in §6 do not change this fp32 result. Dividing fp32 knowledge by hypothetical trit storage would not be a valid comparison.

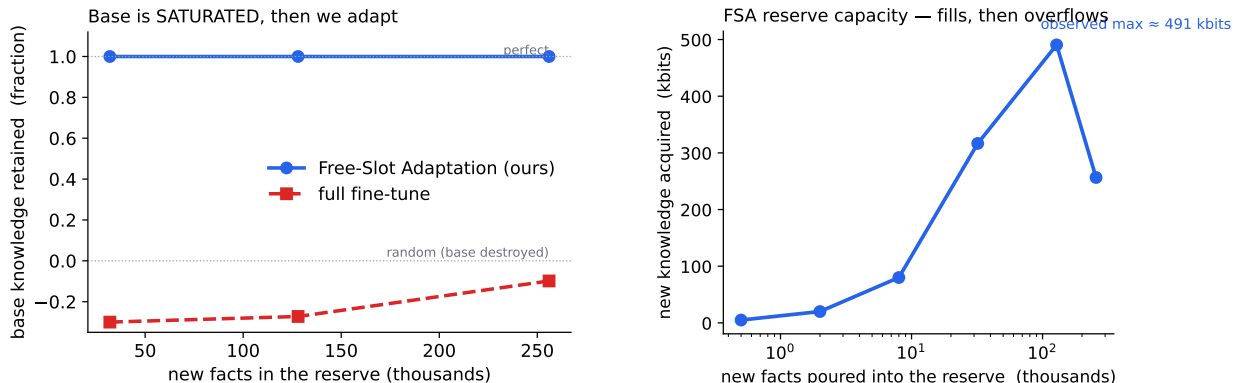


Figure 1: **Left:** masked base retention stays 1.000 (by construction) while full fine-tuning destroys the base. Full-fine-tuning retention is clipped at -0.3 for display; at 32k facts its value is -1.63 . **Right:** the reserve memorizes up to ≈ 490.6 kbits, then *declines* to 256.5 kbits at 256k facts—a maximum under this training budget (fewer exposures per fact as the count grows can contribute), not a proven intrinsic ceiling.

Soft variant (Fig. 3). Run unmasked, retention *degrades* with load ($0.98 \rightarrow 0.78$ at 8k facts $\rightarrow 0.12$ at 32k), bought with full base-input replay. We describe this as degradation, not “graceful” suitability. The λ weight is a plain retention/acquisition dial.

6 Real-model results (lambda 8B)

Model and protocol. Lambda is an 8B-parameter ternary MoE with 128 routed experts per MoE layer, 12 active per token, model width 1152, and a recurrent sequence-mixing state between layers. Each routed expert has a 2304×384 up projection, squared ReLU, and a 384×2304 down projection, stored as **pair8** trits times frozen per-row scales. We patch layers 14–15, all 128 experts, both projections: a broad footprint with 66,807,179 free slots inside already-active pairs. Base support, scales, pair occupancy, router, and all other parameters remain frozen. Plain ternary births take value $\pm\alpha$ for row scale α . The block-scale arm keeps ternary signs but multiplies them by learned magnitudes, one per row and 16 input columns, initialized at

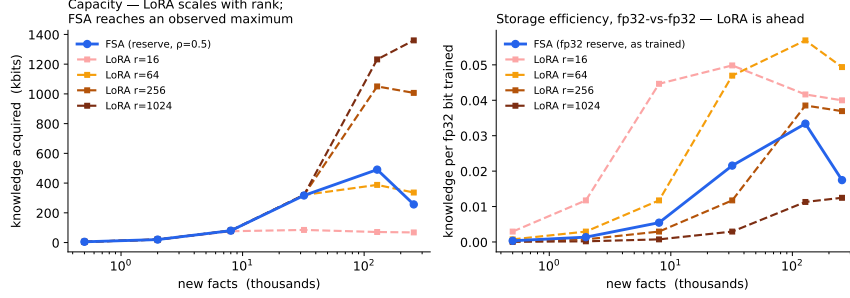


Figure 2: **FSA vs. tuned LoRA.** Left: LoRA’s capacity scales with rank and *exceeds* FSA’s reserve maximum at $r \geq 256$. Right: counted in the representation actually trained (fp32), LoRA is *more* storage-efficient ($\kappa \approx 0.057$ at $r=64$ vs FSA ≈ 0.033). These are uncontrolled dense fp32 mechanism results with different base recalls; the position-inclusive ternary patch comparison on a real model is separate (§6).

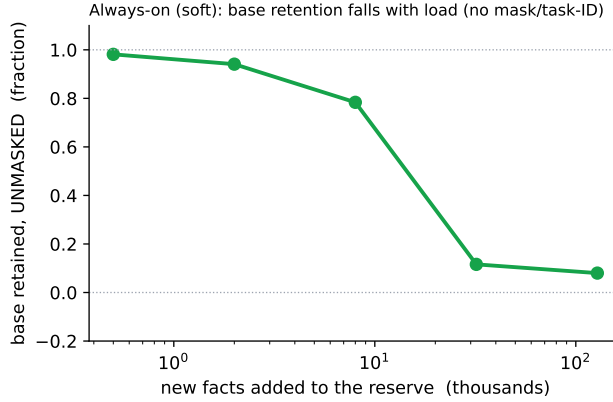


Figure 3: The mask-free soft variant degrades with load (with full base-input replay). It has no addressable slice to delete.

0.5α . Its values leave $\{-\alpha, 0, +\alpha\}$; the packed-format and no-new-parameter claims do not cover this arm. The fp4 variant uses magnitudes $\{0, 0.125, 0.25, 0.375, 0.5, 0.75, 1, 1.5\}\alpha$ (the E2M1 grid scaled by 0.25α), with minimum nonzero magnitude 0.125α and cap 1.5α ; it leaves the ternary codebook. Evaluation decodes experts to bf16. Legal ternary patches could use the same packed kernel, but that serving path was not benchmarked.

The task has 40 synthetic facts about a fictional person (favorite cheese, boat’s name, house number). Training lasts 300 steps with answer-token-only loss: three of 12 phrasing templates are sampled per fact per step, comprising three plain sentences and nine augmented forms, including “Question:/Answer:” and “User:/Assistant:”. Two further phrasings and 16×512 general-text windows provide validation monitoring and checkpoint selection. Test recall is scored under teacher forcing: given the prompt, the model’s greedy (argmax) prediction must match every answer token; termination and extra generated text are not tested. Recall is measured on two phrasings never used in training or validation: “Q: What is X’s attr? A:” and “A little-known fact about X: her attr is”. The Q/A form is held out verbatim but resembles trained question/answer formats. There are 80 test items; one item is 1.25 percentage points, and the unpatched model gets 0/80. Damage is mean per-token ΔNLL relative to the base on 40 disjoint 512-token WikiText windows (about 20k tokens), with paired standard error (SE) over windows; base NLL is 2.890. All results are single-seed; Table 2 reports step 300.

Random flips	Layer 2 KL	Layer 15 KL	Layer 28 KL
1	0.027	0.023	0.007
1,000	0.027	0.023	0.015
1,000,000	0.028	0.026	0.023

Single change	KL	Top-1 agreement
$\varepsilon \leq 0.01$	0	1.000
$\varepsilon = 0.1$	0.0105	0.978
$\varepsilon = 1$	0.0238	0.939

Table 1: Untrained perturbations (single seed). Top: random free-slot flips in down projections. Bottom: one layer-15 weight changed by $\varepsilon\alpha$; small changes are absorbed by bf16 rounding.

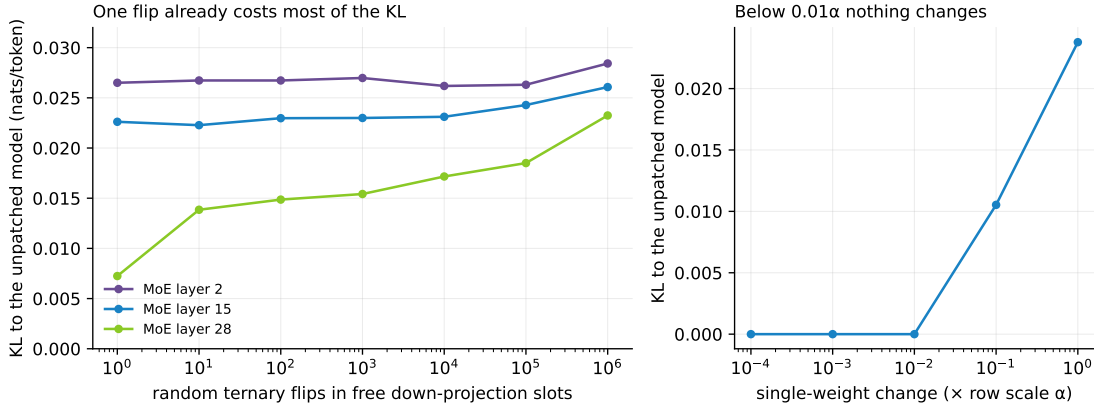


Figure 4: KL against the number of random ternary flips in free down-projection slots at layers 2, 15, and 28 (left), and against a single-weight change relative to its row scale (right). Held-out NLL remains within about 0.01 nats of base (range -0.010 to $+0.005$ relative to 2.890) for the random flips.

6.1 A chaos floor for KL

Random, untrained ternary flips produce a KL floor while held-out NLL stays within about 0.01 nats of the base (range -0.010 to $+0.005$ relative to base NLL 2.890; Table 1). A single layer-15 zero in one expert’s down projection illustrates the bf16 threshold: changes at most 0.01α round away; a full trit changes about 6% of top-1 tokens. Two different single flips disagree with each other by KL 0.027, as much as either with the base, and divergence is flat across sequence position. These observations suggest a different but equally good trajectory after a change exceeds bf16 resolution. Near-ties in discrete MoE routing, propagated through recurrent state, are a plausible driver, not a proven explanation.

For always-on edits the typical KL floor is about 0.02–0.027, with layer-dependent exceptions in Table 1. A KL penalty mostly fights this floor: increasing its weight from 1 to 30 left KL near 0.06 while recall fell from 48% to 20%. We therefore measure damage with paired Δ NLL, interpret KL relative to this floor, and use a local output penalty for preservation (§6.3).

6.2 Recall, damage, and storage

Validation-selected checkpoints do not change the conclusions: Bop $\tau = 0.7$ at step 250 gives 100%/+0.0268, learn-then-prune at 250 gives 90%/+0.0282, and regularized LoRA at 150 gives 95%/+0.0029 on the test recall/damage measures.

Bop’s 80/80 against LoRA’s 78/80 matches, perhaps slightly exceeds, recall: two items on one seed

Method	Test recall	$\Delta\text{NLL} \pm \text{SE}$	Nonzero entries
Ternary Bop, $\tau = 0.7$	100% (80/80)	$+0.0266 \pm 0.0035$	812,420
Ternary Bop, $\tau = 0.9$	100% (80/80)	$+0.0314 \pm 0.0033$	588,673
Ternary learn-then-prune, decay 0.3	91.25% (73/80)	$+0.0250 \pm 0.0032$	542,205
Ternary learn-then-prune, decay 0.1	93.75% (75/80)	$+0.0667 \pm 0.0035$	1,857,504
Signs + learned block scales (16), decay 0.3	87.5% (70/80)	$+0.0282 \pm 0.0031$	1,125,769
Fp4 + local penalty $\lambda = 10$, decay 0.1	96.25% (77/80)	$+0.0050 \pm 0.0015$	21,265,606
LoRA $r = 16$ + local penalty $\lambda = 100$	97.5% (78/80)	$+0.0005 \pm 0.0018$	22,020,096
LoRA $r = 16$, unregularized	97.5% (78/80)	$+0.936 \pm 0.030$	22,020,096

Table 2: Single-seed final checkpoints: 40 facts, layers 14–15, all experts, up and down, answer-only loss, 12 formats. Every run revokes bit-exactly by switching the patch or adapter off and comparing logits to the unpatched model. FSA counts slots; LoRA counts fp32 parameters as trained.

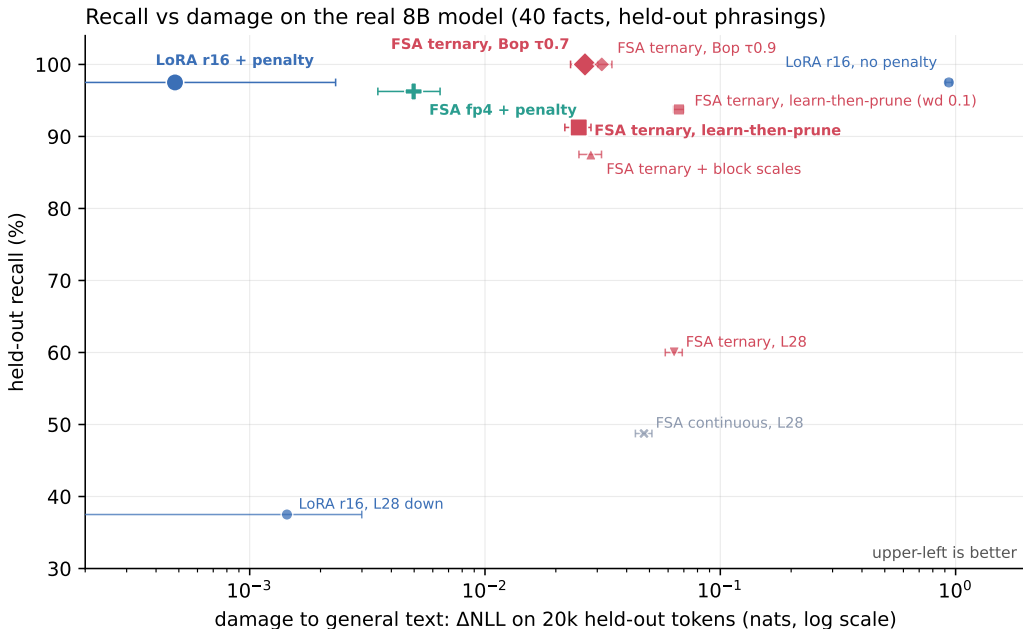


Figure 5: Held-out recall against general-text ΔNLL (log horizontal axis, paired-SE bars). Error bars extending below zero are truncated on the log axis; LoRA + penalty’s interval includes zero. Protocol as in Table 2: answer-only loss, 12 augmented formats, recall on two held-out phrasings (one Q/A), ΔNLL on 40 disjoint 512-token WikiText windows. Besides Table 2, late-layer arms use layer 28, down only, with a 2M-slot budget: gain-minus-cost ternary selection gives 60%/+0.064, continuous bf16 slots 48.75%/+0.047, and same-placement regularized LoRA 37.5%/+0.0014. All are single-seed.

do not establish a significant win. LoRA with the local penalty has damage indistinguishable from zero ($+0.0005 \pm 0.0018$); Bop costs about $+0.027$ nats/token (about 2.7% perplexity), roughly 50 times LoRA’s point estimate, although that estimate is near zero. Fp4 is closest to LoRA on both axes (96.25%, $+0.005$), but leaves the ternary format and uses a much larger patch. Unregularized LoRA is far more damaging ($+0.94$) than any FSA run. Plain ternary writers achieve their moderate damage without a penalty term; the ternary magnitude pin bounds how far a slot can move.

Patch	Entropy-coded positions + values	Indexed positions + values
Bop $\tau = 0.7$, 812,420 slots	0.9 MB	2.7 MB
Bop $\tau = 0.9$, 588,673 slots	0.7 MB	2.0 MB
Learn-then-prune, 542,205 slots	0.6 MB	1.8 MB
Fp4 + penalty, 21.3M slots	18 MB	80 MB
Block-scale run, 1,125,769 slots	1.2 MB + block scales	3.8 MB + block scales

Table 3: Position-inclusive storage estimates. Ternary indexed storage uses a 26-bit index per slot plus its sign. Block scales (one per row and 16 input columns) add about 113 MB in fp32 as implemented, larger than the adapter. LoRA has 22,020,096 parameters: 44 MB in bf16, 88 MB in fp32 as trained.

Bop $\tau = 0.7$ is about 15–50 times smaller (16–49 times using unrounded estimates) than the 44 MB bf16 adapter, depending on position coding (Table 3). Across the plain ternary patches the ratios range about 7–69 times, with learn-then-prune decay 0.1 giving the lowest ratio and decay 0.3 the highest; fp4 at 18–80 MB is not a small patch. Merged plain ternary FSA adds no parameters, tensors, or compute, retaining legal packed values; revocation requires the diff or original pack. Unmerged LoRA adds a low-rank matrix multiplication per patched expert; merged LoRA requires dense bf16 expert weights and forfeits the ternary packed format. On this fact task, regularized LoRA remains the lower-damage method; ternary FSA trades measurable damage for matching recall, a smaller patch, unchanged inference structure, and exact revoke. Multi-session concepts remain untested.

On isolated decoded experts with a synthetic random-fact probe, the ternary reserve learned about 500 facts per expert at about 100% accuracy and revoked bit-exactly, but realized only 0.03–0.07 knowledge bits per slot versus 1.1–1.4 for float slots. In a separate comparison at about 2,000 facts, the full ternary reserve (all structural zeros, requiring a sidecar outside the codebook) had 8–20 times lower weight-level interference than LoRA $r=16/r=64$ under the same drift penalty at matched knowledge. That metric is isolated output drift, distinct from end-to-end general-text damage; on the full model, LoRA with a penalty is better.

6.3 Writers

Bop, latent-free ternary. Extending Helwegen et al. [2019] from binary to ternary, each free slot holds $q_j \in \{-1, 0, +1\}$ directly, with no latent real weight. For loss gradient g_j , maintain

$$m_j \leftarrow (1 - \gamma)m_j + \gamma g_j, \quad \gamma = 0.05.$$

If $q_j = 0$ and $|m_j| > \tau$, birth sets $q_j \leftarrow -\text{sign}(m_j)$. If $q_j \neq 0$, $|m_j| > \tau$, and $\text{sign}(m_j) = q_j$, death sets $q_j \leftarrow 0$: the gradient says the current sign hurts. Reset m_j to zero after each change. Thresholds are 0.7 or 0.9 times the RMS of the first answer-loss gradient over free slots, with no penalty or decay. The threshold controls flips: small gradients need not become full $\pm\alpha$ steps, unlike Adam’s per-coordinate normalization, which can turn persistent small gradients into full flips. The slots are filled early: 812,077 slots were filled at step 50 (812,420 at step 300) at $\tau = 0.7$. Higher τ uses fewer slots (589k) at slightly higher damage, so it is not a simple damage dial.

Learn-then-prune. Each free slot has an fp32 master w_j in row-scale units. Forward uses $q_j = \text{sign}(w_j)\text{I}(|w_j| > 0.5)$; where I is the indicator; gradients pass straight through [Bengio et al., 2013]. AdamW [Loshchilov and Hutter, 2019], learning rate 0.02, applies decoupled weight decay 0.3 to masters. The model learns early: at step 50 validation recall is 97.5%, with 20.3M filled slots and validation $\Delta\text{NLL} +2.2$. Decay pulls weakly supported masters under threshold: 3.2M slots at step 100, 1.1M at 150, and 0.54M at 300. Validation damage falls to about +0.02 and validation recall stays 100%; test recall is separately 91.25%. Held-out damage fell from +0.0282 at step 250 to +0.0250 at step 300, while validation damage rose slightly (+0.016 $\rightarrow$ +0.018). Whether longer pruning helps is untested.

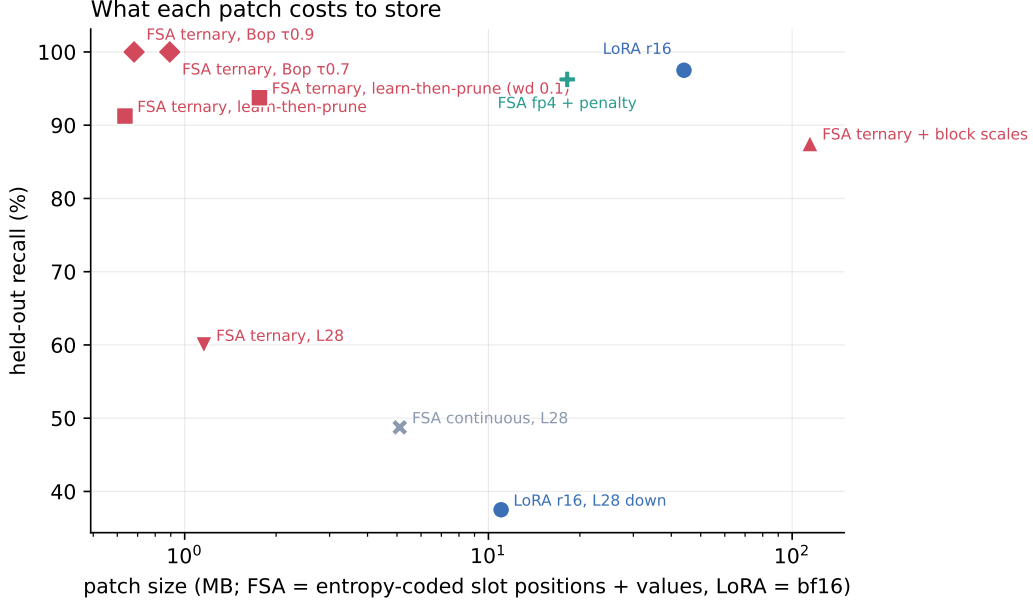


Figure 6: Recall against patch size. Sizes count slot positions, not only values: FSA is entropy-coded positions plus values; LoRA counts bf16 parameters. The plot includes the learned fp32 block scales, bringing that run to about 114 MB, larger than the adapter.

Fp4 slots and local output penalty. Four-bit float slots times α use AdamW decay 0.1 and add the general-text penalty

$$\lambda \frac{\sum_{l,t} \|x_t \Delta W_l^T\|^2}{\sum_{l,t} \|x_t W_l^T\|^2}, \quad \lambda = 10,$$

where l ranges over patched linear projections, t over general-text tokens, x_t is the input to that projection, W_l is its frozen base weight, and ΔW_l is the patch. The denominator, base-weight output energy on the current inputs, is treated as a constant normalizer and is not differentiated. This smooth local penalty compares each patched projection’s direct output change rather than final logits, so it avoids the chaos floor’s whole-model divergence. Its inputs still come from the patched forward pass: earlier patched layers can change later inputs and routing. The best LoRA run uses the same penalty with $\lambda = 100$. For ternary slots it is ineffective: gradient is zero at zero, yet one birth costs a full $\alpha^2 \mathbb{E}[x^2]$; at the strength LoRA needs, it stops ternary learning entirely.

Ingredients that mattered. Answer-only loss avoids learning template boilerplate, which caused most early damage with whole-sentence loss. Twelve diverse formats, including Q/A, closed the validation-to-test gap for both FSA and LoRA (an earlier protocol reached only 54%/56% held-out recall). The broad footprint mattered: 32 selected experts in one layer, down only, with 500k slots starved capacity (19–29% recall). Damage uses a large held-out pool, with checkpoint selection on a separate validation pool.

What did not work. KL penalties fight the chaos floor and cost recall. Ternary slots with smooth penalty and decay peak in recall, then are pruned away. Tiny fp4 slots in late layer 28, down only, with a 2M-slot budget and grid magnitudes starting at 0.03125α , fill about 2.0M slots but yield 0% recall at $+0.0067$ damage: the patch changed the model but learned none of the tested answers. Patch-only learned block scales are no better than plain ternary (87.5%/+0.028) and add substantial storage. Single-late-layer continuous bf16 slots give 48.75%/+0.047, versus same-placement LoRA’s 37.5%/+0.0014: a recall–damage trade, not an overall improvement over LoRA. Gain-minus-cost ternary slot selection there gives 60%/+0.064.

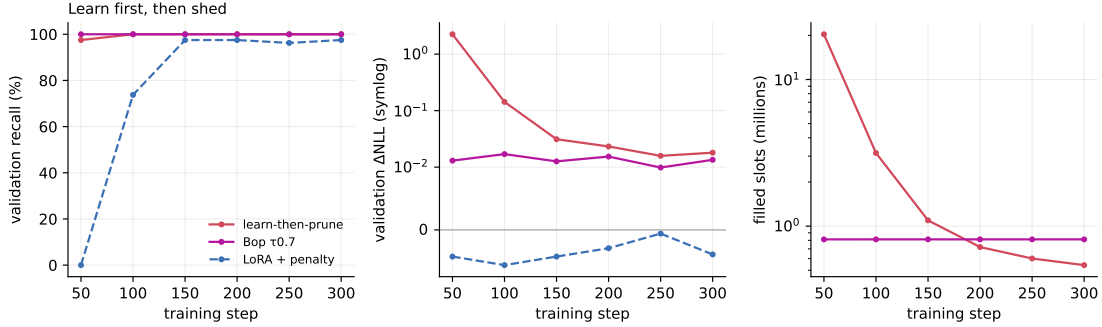


Figure 7: Validation recall, validation ΔNLL (symmetric log scale), and filled slots (millions, log scale) across training. The recall and damage panels show learn-then-prune, Bop $\tau = 0.7$, and regularized LoRA; the filled-slot panel shows only the two FSA writers. Bop’s filled-slot count is set by its threshold early and then stays flat, while learn-then-prune fills many slots first and sheds them over training. Validation trajectories differ from the held-out test endpoints in Table 2.

7 A proposed direction (untested): multi-session learning

The ternary reserve now carries facts on a real model (§6). A revocable domain patch is therefore demonstrated for this task: enable the slice for its domain; disable it to recover the base. The personal-model application remains an untested proposal, with a narrower contract:

Patch-revocation contract. Deleting an isolated patch exactly removes *that patch’s parameter contribution*. Equivalence to a model trained without its data additionally requires that no retained patch or training state depended on it. It does not delete individual facts (facts within a slice are superimposed), and it does not make information already in the base inaccessible.

Retrieval deletes individual records while leaving the base unchanged [Lewis et al., 2020, Khandelwal et al., 2020]; disabling a per-user adapter removes that user’s contribution [Tan et al., 2024]. The case for weights is skills, style, and policy that benefit from adaptation in every forward pass. Arbitrary facts remain a retrieval baseline’s natural task. The synthetic fp32 assay favors tuned LoRA on storage; the real ternary fact experiment offers smaller patches at greater damage, not a general personalization result.

The pre-registered EP-1 experiment is in progress and explicitly **untested**: five sessions with a synthetic user containing 40 policies, recurrent facts, and one-off “lure” facts, consolidating self-study into free ternary slots. An identifiability gate requires the in-context ceiling to beat the unpatched model by at least 10 concept points on development probes; otherwise nothing is scored. Success criteria fixed in advance require, after session 5, concept gain of at least 10 percentage points over the unpatched model on held-out probes, lure recall no more than 5 points above it, and mean general-text ΔNLL at most $+0.01$ nats with a paired 95% damage interval excluding $+0.05$. Final concept accuracy must retain at least 80% of the best earlier session’s accuracy, with no upward ratchet in lure recall (monotone non-decreasing with a net increase).

The pre-registered revoke criterion requires every patched weight tensor to be bitwise equal to the base and format-legal, logits on 32 fixed prompts within 10^{-2} of the base, and concept/lure metrics back to the base within noise. Exact logit equality is reported separately; it was observed when switching patches off in the fact experiment (§6). The comparator is regularized LoRA at the same placement. “Match LoRA” means within 3 concept points and no more than $+0.005$ extra damage, with the lure constraint held and revoke passing; “beat LoRA” means at least 5

more concept points with damage no worse than LoRA. One seed is a pilot. Neither multi-session concept learning nor on-device training is established: optimizer state can be large even when a final patch is small. A broader evaluation should include local retrieval and per-user adapters on personalization benchmarks [Salemi et al., 2024], with unlearning measured [Bourtole et al., 2021, Maini et al., 2024].

8 Limitations and the remaining experiments

Gate 0 is partly answered. The real ternary reserve, trained through legal format values on a real model with frozen occupancy and scales, carries the facts and has a position-inclusive storage advantage (§6). This does not settle damage: Bop’s +0.027 nats exceeds the EP-1 +0.01 target, and regularized LoRA is better. Nor does it settle the multi-session concept task. The decisive next evidence is that pre-registered experiment and whether ternary writers can lower damage without losing recall. Longer pruning and packed-kernel serving remain untested.

All numbers are single-seed; the 40-fact test has only 80 items, its Q/A form resembles training, and the roughly 20k-token damage pool cannot establish broad behavioral safety. The synthetic fp32 comparison is uncontrolled, while real-model writer and regularization choices also differ. The chaos-floor explanation is a hypothesis. K measures log-loss improvement, not semantic knowledge; knowledge efficiency is assay- and load-specific, and from-scratch quantization results cannot predict frozen-reserve capacity. The frozen base can augment or suppress through new paths but cannot refine its existing weights. Exact patch-off restoration does not imply preservation while a patch is on, per-fact deletion, or removal of knowledge already in the base.

9 Conclusion

The synthetic fp32 study establishes a freezing mechanism with exact masked retention, bounded memorization, and a degrading always-on variant; its tuned LoRA comparison remains unfavorable to FSA. On a real 8B ternary MoE, Bop recalls 80/80 held-out items from a 0.9 MB position-inclusive patch, with exact revoke and preserved packed values, at about +0.027 nats/token damage. Regularized LoRA recalls 78/80 with no measurable damage: matching recall and smaller storage do not establish an overall FSA win. Damage is the open problem. The pre-registered multi-session concept experiment is in progress; a private model that learns across sessions remains an untested proposal.

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